ASSIGNMENT 04

1. **Declare function and call it by reference for swapping numbers**

Code:-

```
#include<iostream>
using namespace std;

void noSwap(int &n1, int &n2){
	int tmp = n1;
	n1=n2;
	n2=tmp;
}

int main(){
	int num1, num2;
	cout<<"Enter 2 nos: "<<endl;
	cin>>num1>>num2;
	cout<<"Before swap: "<<endl;
	cout<<"num1: "<<num1<<" Num2: "<<num2<<endl;
	noSwap(num1, num2);
	cout<<"After swap: "<<endl;
	cout<<"num1: "<<num1<<" Num2: "<<num2<<endl;
	return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$ ./a.out
Enter 2 nos:
22
44
Before swap:
num1: 22 Num2: 44
After swap:
num1: 44 Num2: 22
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$
```

2. **Create function for printing an array and call it from main (Student array)**

Code:-

```
#include<iostream>
#include<cstring>
using namespace std;

void addStudent(char** std, int size){
    for(int i=0; i<size; i++){
        cout<<"Enter "<<i+1<<" Student Name: "<<endl;
        cin>>std[i];
    }
}

void displayStudent(char** std, int size){
    for(int i=0; i<size; i++){
        cout<<std[i]<<endl;
    }
}

int main(){
    int size;
    cout<<"Enter size of array"<<endl;
    cin>>size;
    char** student = new char*[size];
    for(int i=0; i<size; i++){
        student[i] = new char[100];
    }
    addStudent(student, size);
    displayStudent(student, size);
    return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$ ./a.out
Enter size of array
3
Enter 1 Student Name:
Sanket
Enter 2 Student Name:
Shubham
Enter 3 Student Name:
Tejas
Sanket
Shubham
Tejas
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$
```

3. **Function for print factorial series with recursive**

Code:-

```
#include<iostream>
using namespace std;

int factNum(int n){
    if(n==0){
        return 1;
    }
    return n*factNum(n-1);
}
int main(){
    int num;
    cout<<"Enter number: "<<endl;
    cin>>num;
    for(int i=0; i<=num; i++){
        cout<<factNum(i)<<"\t";
    }
    cout<<endl;
    return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$ g++ factRecursive.cpp
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$ ./a.out
Enter number:
5
1       1       2       6       24      120
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$
```

4. **Reverse string (char array) with recursion**

Code:-

```
#include<iostream>
#include<cstring>
using namespace std;

void reverseStr(char str[]){
    if(*str == '\0' ){
        return;
    }
    reverseStr(str+1);
    cout<<*str;
}

int main(){
    char str[20];
    cout<<"Enter a string: "<<endl;
    cin>>str;
    //int len = strlen(str);
    reverseStr(str);
    cout<<endl;
    return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$ g++ reverseString.cpp
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$ ./a.out
Enter a string:
thisisstring
gnirtssisiht
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$
```

5. **Create Student class and add rollNo and name in it. Create Accept() func-tion which will take an array of Students as input. Write one more func-tion to display student data. This Display() function should receive Stu-dent array.**

Code:-

```
#include<iostream>
#include<cstring>
using namespace std;

class Student{
   public:
      int rollNo;
      string name;
};

void Acceptdata(Student s[], int size){
   cout<<"Enter student details: "<<endl;
   for(int i=0; i<size; i++){
      cout<<"Enter roll no: "<<i+1<<" student"<<endl;
      cin>>s[i].rollNo;
      cout<<"Enter name: "<<i+1<<" student"<<endl;;
      cin>>s[i].name;
   }

}

void Displaydata(Student s[], int size){
   for(int i=0; i<size; i++){
      cout<<"Roll no: "<<s[i].rollNo<<endl;
      cout<<"Name: "<<s[i].name<<endl;
   }
}

int main(){
   int size;
   cout<<"Enter size of array: "<<endl;
   cin>>size;
   Student s[size];
```

```
    Acceptdata(s, size);
    Displaydata(s, size);
    return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$ g++ Studentclass.cpp
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$ ./a.out
Enter size of array:
3
Enter student details:
Enter roll no: 1 student
101
Enter name: 1 student
sanket
Enter roll no: 2 student
102
Enter name: 2 student
shubham
Enter roll no: 3 student
3
Enter name: 3 student
tejas
Roll no: 101
Name: sanket
Roll no: 102
Name: shubham
Roll no: 3
Name: tejas
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$
```

6. **Menus:**
   a) Accept n numbers from users and store them in to dynamically allocated array. Write a function to sort and display sorted array.
   b) Accept n characters from user and print is as string.
   c) Accept n numbers from user &write function to find average and display average.
   d) Accept n numbers from user and write a function to multiply each number by 2 And display result.

Code:-

```
//Menu driven program

#include<iostream>
using namespace std;
void acceptNum(int* arr, int n){        //function for accepting number array

   cout<<"Enter elements: "<<endl;
   for(int i=0; i<n; i++)
   {
      cin>>arr[i];
   }
}

void acceptChar(char* charArray, int n){    //function for accepting character array

   cout<<"Enter Character: "<<endl;
   for(int i=0; i<n; i++)
   {
      cin>>charArray[i];
   }
}

void displayNum(int* arr, int n){       //function for displaying number array
   for(int i=0; i<n; i++)
   {
      cout<<arr[i]<<"  ";
   }
   cout<<endl;
}

void displayChar(char* charArray, int n){       //function for displaying character array
```

```
    for(int i=0; charArray[i]!='\0'; i++)
    {
        cout<<charArray[i];
    }
    cout<<endl;
}

void sort(int* arr, int n){             //function for sorting number array
    int  tmp;
    for(int i=0; i<n; i++){
        for(int j=0; j<n-1; j++)
        {
            if(arr[j]>arr[j+1]){
                tmp = arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=tmp;
            }
        }
    }
}
void average(int* arr, int n){      //function for finding average of array elements
    int sum=0;
    for(int i=0; i<n; i++){
        sum+=arr[i];
    }
    cout<<"Average of input number is: "<<(sum/n)<<endl;
}

void multiby2(int* arr,int n){      //function for multiplying number array by 2
    for(int i=0; i<n; i++){
        arr[i]=arr[i]*2;
    }
}

int main(){
    int n;
    int* arr = new int[n];
    char* charArray = new char[n];
    int choice;
    do{
```

```
    cout<<"********** Menu **********"<<endl;
    cout<<"1. sorted array\n2. print string\n3.find average\n4. multiply by 2\n5.exit"<<endl;
    cout<<"Enter your choice:"<<endl;
    cin>>choice;

    switch(choice){
        case 1: cout<<"Enter Total number of elements: "<<endl;
                cin>>n;
                acceptNum(arr, n);
                sort(arr, n);
                displayNum(arr, n);
                break;

        case 2: cout<<"Enter Total number of elements: "<<endl;
                cin>>n;
                acceptChar(charArray, n);
                displayChar(charArray, n);
                break;


        case 3: cout<<"Enter Total number of elements: "<<endl;
                cin>>n;
                acceptNum(arr, n);
                average(arr, n);
                //displayNum(arr, n);
                break;

        case 4: cout<<"Enter Total number of elements: "<<endl;
                cin>>n;
                acceptNum(arr, n);
                multiby2(arr, n);
                displayNum(arr, n);
                break;
        case 5: //exit
                cout<<"Exiting....."<<endl;
                break;
        default: cout<<"Invalid Input!!!"<<endl;
                break;
    }
}while(choice!=5);
```

```
    if(charArray!=NULL){
        delete[] charArray;
    }
    if(arr!=NULL){
        delete[] arr;
    }
    return 0;
}
```

O/P:-

```
pgcp-ac@dac:~/Batch2/cpp day 4/Assignments$ ./a.out
********** Menu **********
1. sorted array
2. print string
3.find average
4. multiply by 2
5.exit
Enter your choice:
1
Enter Total number of elements:
5
Enter elements:
11
1
45
23
09
1  9  11  23  45
********** Menu **********
1. sorted array
2. print string
3.find average
4. multiply by 2
5.exit
Enter your choice:
2
Enter Total number of elements:
6
Enter Character:
s
a
n
k
e
t
sanket
```

```
********** Menu **********
1. sorted array
2. print string
3.find average
4. multiply by 2
5.exit
Enter your choice:
3
Enter Total number of elements:
5
Enter elements:
1
2
3
4
5
Average of input number is: 3
********** Menu **********
1. sorted array
2. print string
3.find average
4. multiply by 2
5.exit
Enter your choice:
4
Enter Total number of elements:
5
Enter elements:
10
20
30
40
50
20  40  60  80  100
```